

CANDIDATE
NAME

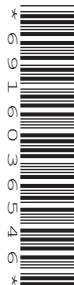
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CENTRE
NUMBER

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9231/11

October/November 2020

2 hours

You will need: List of formulae (MF19)

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [].

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[Turn over

- State the type of each transformation, and make clear the order in which they are applied. [2]

[illegible]

(b) Find, in terms of a and b , the matrix which transforms parallelogram $OPQR$ onto the unit square. [2]

[illegible]

[illegible]

- 2 (a) Use standard results from the List of Formulae (MF19) to show that

$$\sum_{r=1}^n (7r+1)(7r+8) = an^3 + bn^2 + cn,$$

where a , b and c are constants to be determined. [3]

[illegible]

- (b) Use the method of differences to find $\sum_{r=1}^n \frac{1}{(7r+1)(7r+8)}$ in terms of n . [4]

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- (c) Deduce the value of $\sum_{r=1}^{\infty} \frac{1}{(7r+1)(7r+8)}$. [1]

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3 The cubic equation $x^3 + cx + 1 = 0$, where c is a constant, has roots α, β, γ .

(a) Find a cubic equation whose roots are $\alpha^3, \beta^3, \gamma^3$. [3]

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(b) Show that $\alpha^6 + \beta^6 + \gamma^6 = 3 - 2c^3$. [3]

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- (c) Find the real value of c for which the matrix $\begin{pmatrix} 1 & \alpha^3 & \beta^3 \\ \alpha^3 & 1 & \gamma^3 \\ \beta^3 & \gamma^3 & 1 \end{pmatrix}$ is singular. [5]

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

(c) Find the acute angle between the planes OAB and ABC . [4]

[illegible]

- 5** Prove by mathematical induction that, for every positive integer n ,

$$\frac{d^{2n-1}}{dx^{2n-1}}(x \sin x) = (-1)^{n-1}(x \cos x + (2n-1) \sin x). \quad [7]$$

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

- 6** The curve C has equation $y = \frac{x^2 + x - 1}{x - 1}$.

(a) Find the equations of the asymptotes of C .

[3]

This image shows a single sheet of white paper with ten horizontal dashed lines, typical of primary-ruled notebook paper. The lines are evenly spaced and extend across the width of the page. There is no handwriting or other markings on the paper.

(b) Show that there is no point on C for which $1 < y < 5$.

[4]

[illegible]

- (c) Find the coordinates of the intersections of C with the axes, and sketch C . [3]

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- (d) Sketch the curve with equation $y = \left| \frac{x^2 + x - 1}{x - 1} \right|$. [2]

has polar equation $r = \sin 4\theta$. [4]

[illegible]

The curve C has polar equation $r = \sin 4\theta$, for $0 \leq \theta \leq \frac{1}{4}\pi$.

- (b) Sketch C and state the equation of the line of symmetry. [3]

- (c) Find the exact value of the area of the region enclosed by C . [4]

[illegible]

- (d) Using the identity $\sin 4\theta \equiv 4 \sin \theta \cos^3 \theta - 4 \sin^3 \theta \cos \theta$, find the maximum distance of C from the line $\theta = \frac{1}{2}\pi$. Give your answer correct to 2 decimal places. [6]

[illegible]

[illegible]

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